



Artificial Intelligence (AI-2002)

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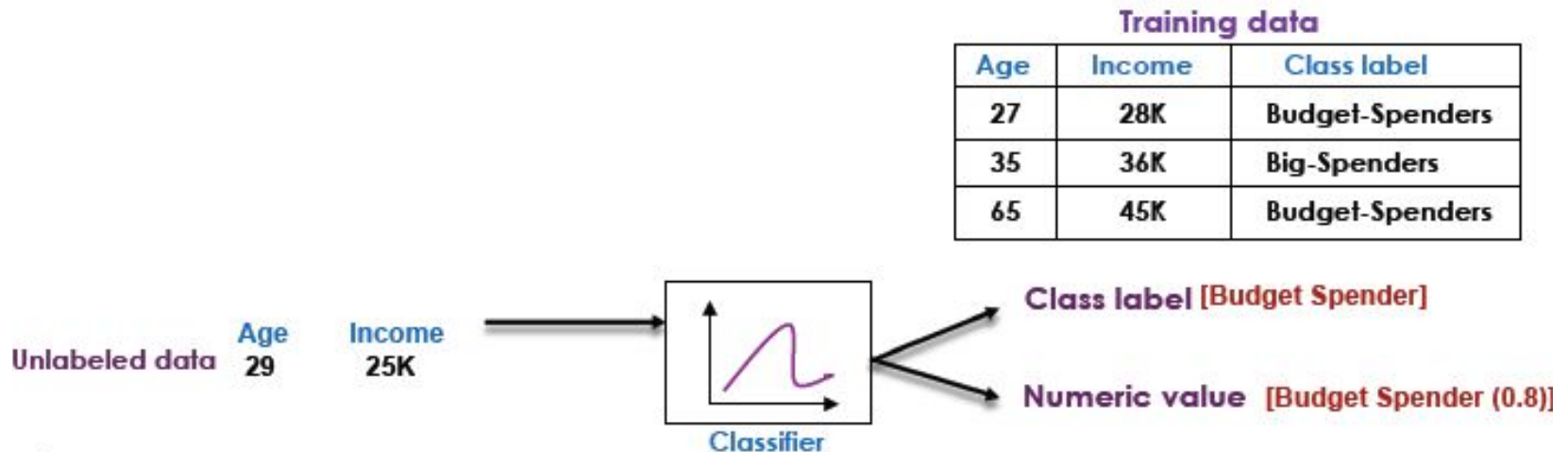
ML- Classification and Regression

Definition

❖ Classification is also called **Supervised Learning**

❖ **Supervision**

- The training data (observations, measurements, etc) are used to learn a classifier
- The training data are **labeled data**
- New data (**unlabeled**) are **classified** Using the training data



❖ **Principle**

- Construct models (functions) based on some training examples
- Describe and distinguish classes or concepts for future prediction
- Predict some unknown class labels

Supervised vs. Unsupervised Learning

❖ Supervised learning (classification)

- Supervision: The training data (observations, measurements, etc.) are accompanied by labels indicating the class of the observations
- New data is classified based on the training set

❖ Unsupervised learning (clustering)

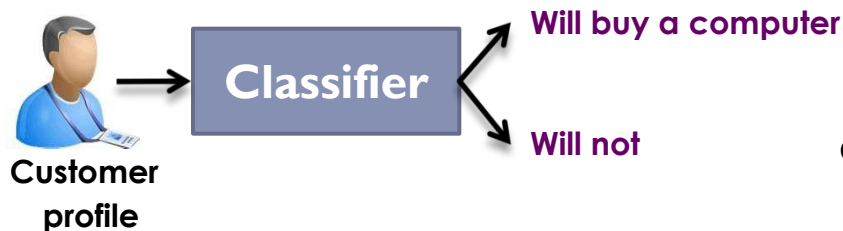
- The class labels of training data is unknown
- Given a set of measurements, observations, etc. with the aim of establishing the existence of classes or clusters in the data

Classification vs. Prediction

Classification

- Predicts categorical class labels (discrete or nominal)
- Use labels of the training data to classify new data

Example



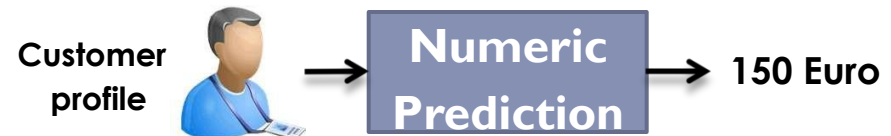
- A model or classifier is constructed to predict **categorical labels** such as “safe” or “risky” for a loan application data.

Prediction

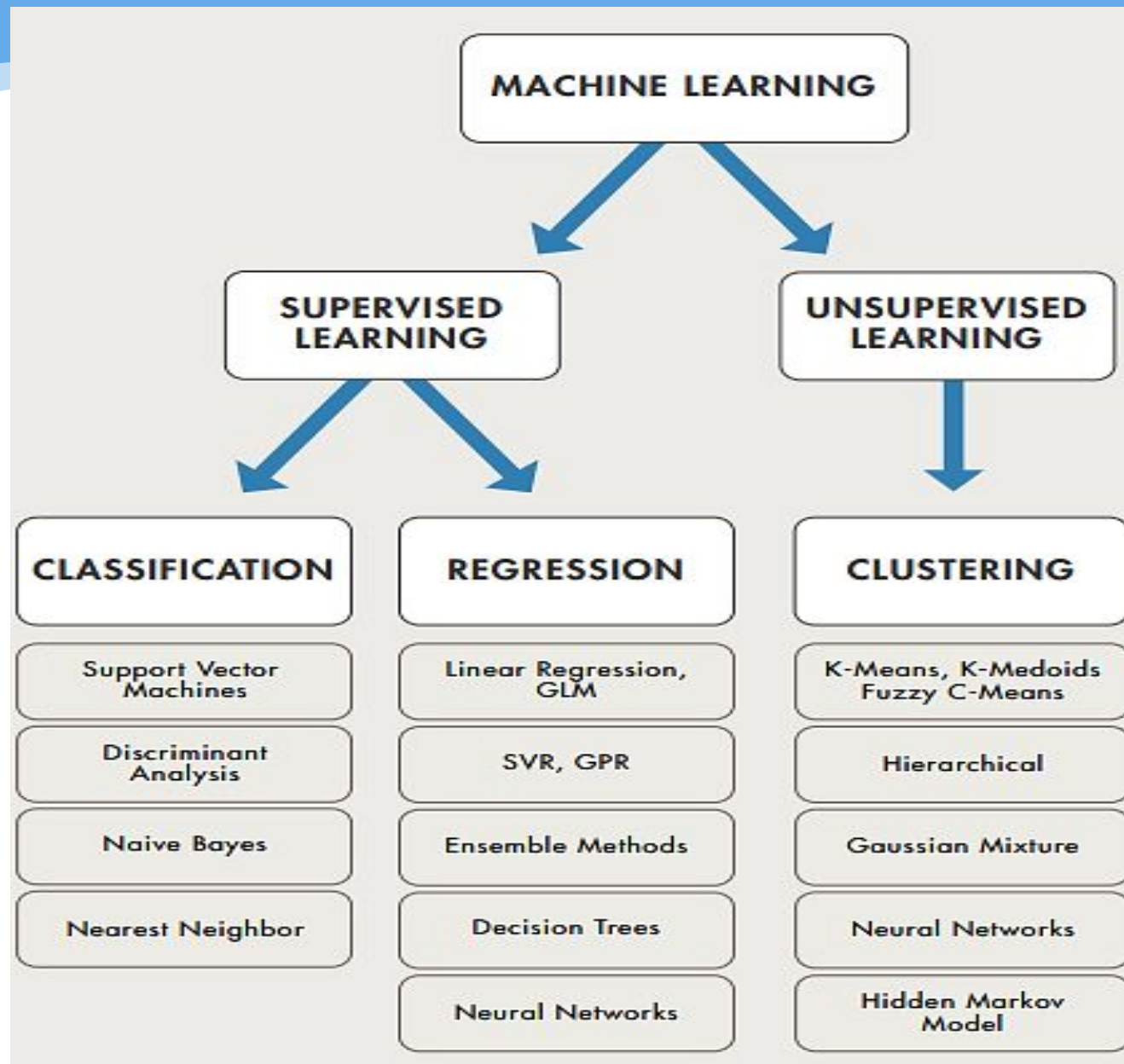
- ❖ Models continuous-valued functions, i.e., predicts unknown or missing values

Example

- ❖ A marketing manager would like to predict how much a given customer will spend during a sale



- ❖ Unlike classification, it provides ordered values
- ❖ **Regression analysis is used for prediction**
- ❖ Prediction is a short name for **numeric prediction**



Applications

❖ Typical applications

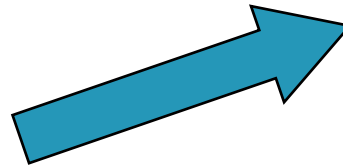
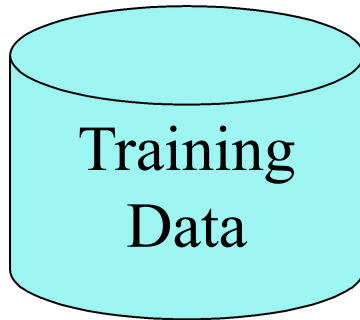
- Credit approval
- Target marketing
- Medical diagnosis
- Treatment effectiveness analysis
- Fraud detection
- Defective parts identification in manufacturing
- Crime zoning



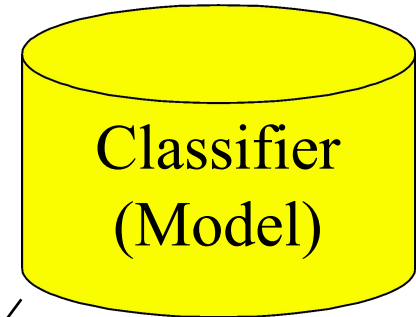
Classification—A Three-Step Process

- ❖ **Model construction:** describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- ❖ **Test Classifier (Estimate accuracy of the model)**
 - The known label of **test sample** is compared with the classified result from the model
 - Accuracy rate is the **percentage** of test set samples that are correctly classified by the model
 - Test set is **independent** of training set, otherwise **over-fitting** will occur
- ❖ **Model usage:** for classifying future or unknown objects
 - If the accuracy is acceptable, use the model to **classify data** tuples whose class labels are not known

Process (1): Model Construction



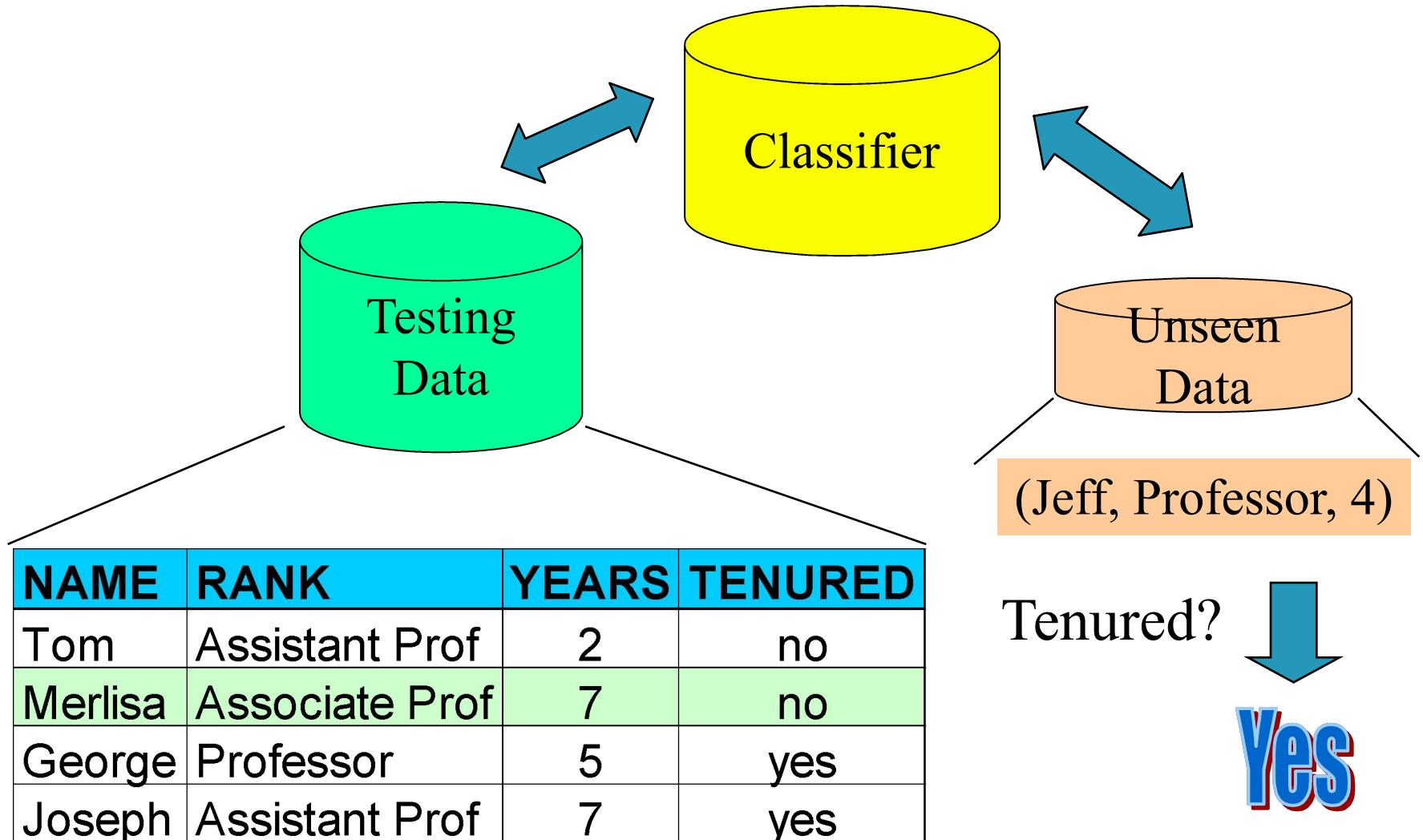
Classification
Algorithms



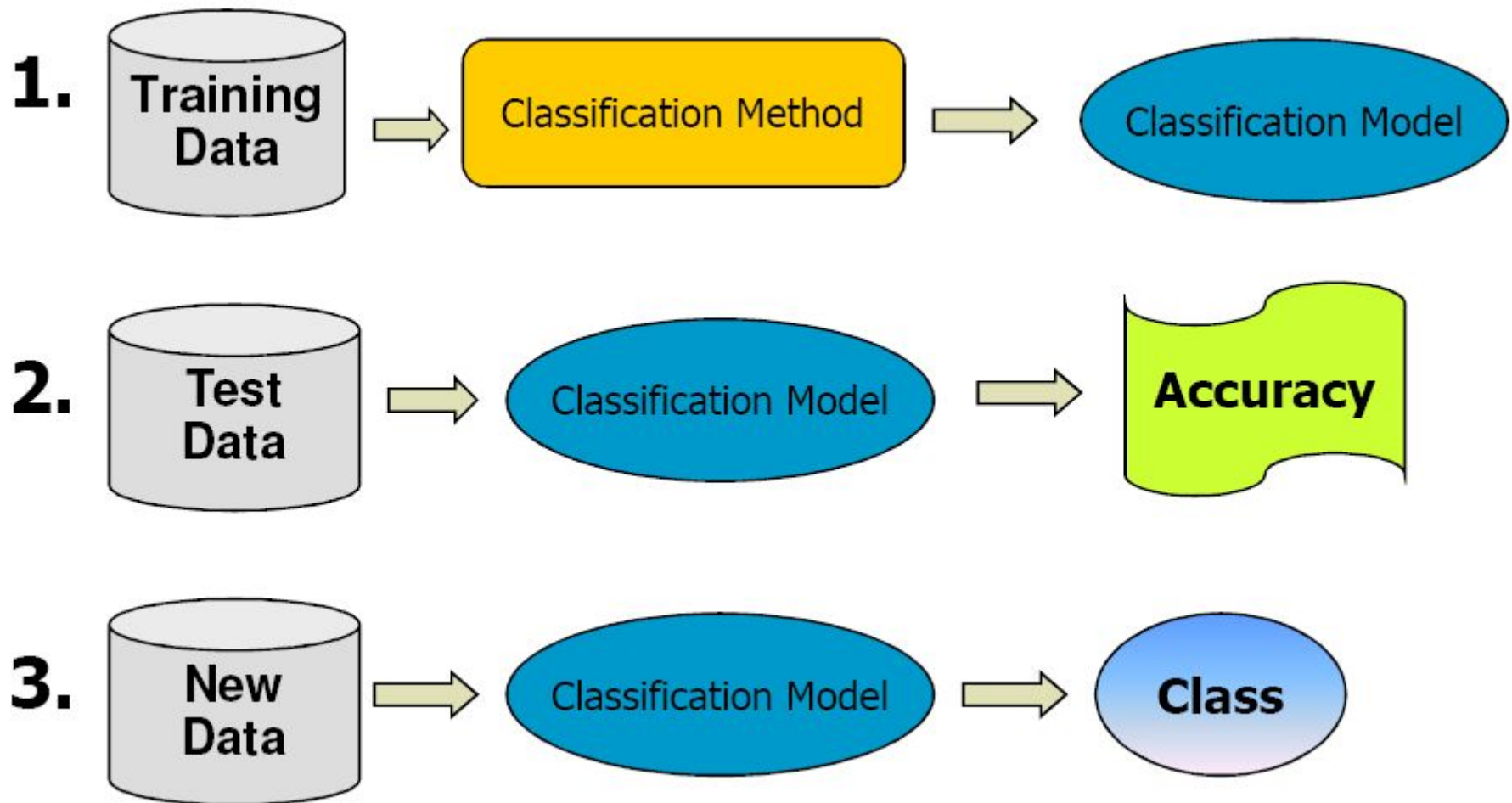
NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

IF rank = 'professor'
OR years > 6
THEN tenured = 'yes'

Process (2&3): Test and Using the Model in Prediction



Classification—A Three-Step Process



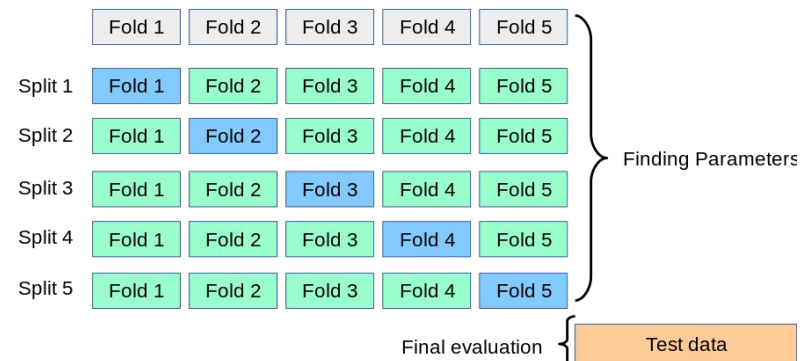
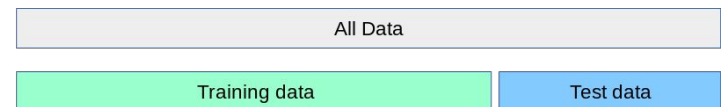
Cross Validation- CV

LOOCV

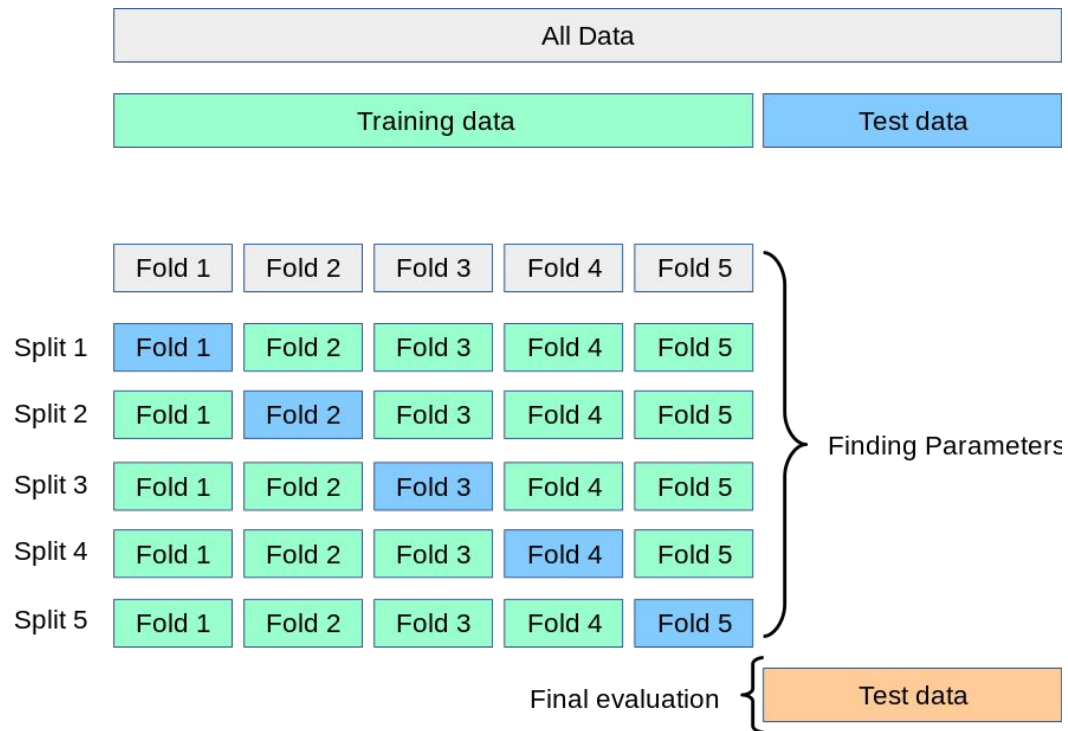
- ❖ Leave-one-out CV is a validation method where **one data point is used for testing and all remaining data points are used for training**, repeated for every sample in the dataset.
- ❖ **LOOCV = Train on (N-1) samples, test on 1 sample, repeat N times.**

K-fold

- ❖ k-Fold CV splits the dataset into **k equal parts (folds)**. Each fold is used **once as the test set**, and the remaining k-1 folds are used for training.
- ❖ Average the errors across all k iterations for the final model performance.



K-fold CV



Trade-off between Bias and Variance

- ❖ **Bias** is the error caused by **too simple assumptions** in the model.
- ❖ The model **cannot capture the true pattern** in data.
- ❖ **Result: Underfitting.**
- ❖ **Bias is Training Errors**
- ❖ **Variance** is the error caused by the model being **too sensitive to training data**.
- ❖ The model learns noise instead of general patterns.
- ❖ **Result: Overfitting.**
- ❖ **Variance is Testing Errors**



Underfitting and Overfitting

- ❖ Underfitting happens when the model is too simple and fails to learn enough meaningful patterns from the data, resulting in higher loss because it has not captured the underlying signal properly.
- ❖ Overfitting occurs when the model becomes too complex and starts learning not only the important patterns but also the noise or random variations in the training data, which also leads to poor performance.
- ❖ The key to building an effective model is to find the right balance between these two, so that the model learns the true patterns without being too simple or too complex.

Ockham's Razor principle

- ❖ In ML, When multiple hypotheses are consistent with the observed data, the simplest one should be preferred.
- ❖ Because simpler models:
 - Generalize better (work well on unseen data).
 - Avoid **overfitting**.
 - Are easier to interpret.

Issues: Data Preparation

- ❖ Data cleaning
 - Preprocess data in order to reduce noise and handle missing values
- ❖ Relevance analysis (feature selection)
 - Remove the irrelevant or redundant attributes
- ❖ Data transformation
 - Generalize and/or normalize data

Bayesian Classification: Why?

- ❖ A **statistical classifier**: performs probabilistic prediction, i.e., predicts class membership probabilities
- ❖ **Foundation**: Based on Bayes' Theorem.
- ❖ **Performance**: A simple Bayesian classifier, naïve Bayesian classifier, has comparable performance with decision tree and selected neural network classifiers
- ❖ **Incremental**: Each training example can incrementally increase/decrease the probability that a hypothesis is correct — prior knowledge can be combined with observed data
- ❖ **Standard**: Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayesian Theorem: Basics

Deep Intuition Behind the Naïve Bayes Classifier

1. Starting Point: Bayes' Theorem

Naïve Bayes is built on **Bayes' Theorem**, which tells us how to update our belief about an event after seeing evidence.

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Meaning of Variables

- $A \rightarrow$ Hypothesis (what we want to predict)
- $B \rightarrow$ Observed evidence (data we see)

So,

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

Where:

- $P(A) \rightarrow$ Prior belief about A

Towards Naïve Bayesian Classifier

2. Moving Toward Classification

In machine learning classification:

- Let y = class label (Spam / Not Spam, Yes / No, etc.)
- Let $X = \{x_1, x_2, x_3, \dots, x_n\}$ = features (attributes)

Example:

- x_1 = Refund
- x_2 = Marital Status
- x_3 = Income
- y = Taxable = Yes/No

Goal:

Predict $P(y \mid x_1, x_2, \dots, x_n)$

Meaning:

👉 *What is the probability of class y given all observed features?*

3. Applying Bayes' Theorem

Using Bayes theorem:

$$P(y \mid x_1, x_2, \dots, x_n) = \frac{P(x_1, x_2, \dots, x_n \mid y) P(y)}{P(x_1, x_2, \dots, x_n)}$$

Interpretation

- $P(y) \rightarrow$ prior probability of class
- $P(x_1, \dots, x_n \mid y) \rightarrow$ probability of observing features if class is y
- Denominator $\rightarrow$ same for all classes (acts as scaling factor)

So for classification we focus on:

$$P(y) P(x_1, x_2, \dots, x_n \mid y)$$

4. The Problem (Why Naïve Bayes Exists)

Computing:

$$P(x_1, x_2, \dots, x_n \mid y)$$

is **extremely hard** because features may depend on each other.

Example:

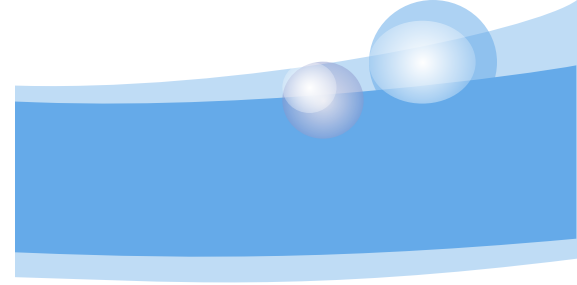
Income and occupation are often related.

Estimating joint probabilities requires huge data:

$$P(x_1, x_2, x_3, \dots, x_n \mid y)$$

➡ becomes exponentially complex.

Derivation of Naïve Bayes Classifier



5. The "Naïve" Assumption (Key Idea)

Naïve Bayes makes one simplifying assumption:

All features are conditionally independent given the class.

Mathematically:

$$x_i \perp x_j \mid y$$

Meaning:

Once we know the class y , features no longer influence each other.

6. The Beautiful Simplification

Because of conditional independence:

$$P(x_1, x_2, \dots, x_n \mid y) = \prod_{i=1}^n P(x_i \mid y)$$

So the classifier becomes:

$$P(y \mid x_1, \dots, x_n) = \frac{P(y) \prod_{i=1}^n P(x_i \mid y)}{P(x_1, \dots, x_n)}$$

Since denominator is constant across classes:

$$P(y \mid X) \propto P(y) \prod_{i=1}^n P(x_i \mid y)$$

7. Final Decision Rule (Classifier)

Naïve Bayes predicts:

$$\hat{y} = \arg \max_y \left[P(y) \prod_{i=1}^n P(x_i \mid y) \right]$$

Naïve Bayesian Classifier: Training Dataset

Class:

C_1 :buys_computer = 'yes'

C_2 :buys_computer = 'no'

Data to be classified:

$X = (\text{age} \leq 30,$
 $\text{income} = \text{medium},$
 $\text{student} = \text{yes},$
 $\text{credit_rating} = \text{fair})$

$X \square \text{buy computer?}$

age	income	student	credit rating	com
≤ 30	high	no	fair	no
≤ 30	high	no	excellent	no
31...40	high	no	fair	yes
> 40	medium	no	fair	yes
> 40	low	yes	fair	yes
> 40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤ 30	medium	no	fair	no
≤ 30	low	yes	fair	yes
> 40	medium	yes	fair	yes
≤ 30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
> 40	medium	no	excellent	no

Naïve Bayesian Classifier: An Example

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

❖ $X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$

❖ Likelihood of “yes” = $\frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14}$
= **0.02821**

❖ Likelihood of “no” = $\frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14}$
= **0.0068**

❖ Turn into probability by normalizing likelihood values
probability of “yes” = $0.02821 / (0.02821 + 0.0068)$
= **0.8057**

❖ probability of “no” = $0.0068 / (0.02821 + 0.0068)$
= **0.1942**

age			Income			Student			Credit Rating			Buys_Computer	
	Yes	No		Yes	No		Yes	No		Yes	No	Yes	No
<=30	2/9	3/5	High	2/9	2/5	Yes	6/9	1/5	Excellent	3/9	3/5	9/14	5/14
31..40	4/9	0/5	Medium	4/9	2/5	No	3/9	4/5	Fair	6/9	2/5		
>40	3/9	2/5	Low	3/9	1/5								

probability of “yes” is 0.8057 therefore X belongs to (“buy_computer = yes”)

Avoiding the Zero-Probability Problem

- ❖ Naïve Bayesian prediction requires each conditional prob. be non-zero. Otherwise, the predicted prob. will be zero

$$P(X | C_i) = \prod_{k=1}^n P(x_k | C_i)$$

- ❖ Ex. Suppose a dataset with 1000 tuples, income=low (0), income=medium (990), and income = high (10),
- ❖ Use Laplacian correction (or **Laplacian estimator**)
 - Adding 1 to each case
 - Prob(income = low) = 1/1003
 - Prob(income = medium) = 991/1003
 - Prob(income = high) = 11/1003
 - The “corrected” prob. estimates are close to their “uncorrected” counterparts

Previous Example

If any one probability becomes 0, the whole product becomes 0

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Class: $P(C) = N_c / N$
 - e.g., $P(\text{No}) = 7/10$,
 $P(\text{Yes}) = 3/10$
- For discrete attributes: $P(A_i | C_k)$
 $= |A_{ik}| / N_c$
 - where $|A_{ik}|$ is number of instances having attribute A_i and belongs to class C_k
 - Examples:
 $P(\text{Status}=\text{Married} | \text{No}) = 4/7$
 $P(\text{Refund}=\text{Yes} | \text{Yes})=0$

Laplace Smoothing

$$P = \frac{\text{count} + 1}{\text{total} + k}$$

❖ Where:

- **count** = how many times feature occurs with class
- **total** = total instances of that class
- **k** = number of possible values of that feature

❖ P(Refund = Yes | Evade = Yes)

❖ From dataset:

- count = 0
- total = 3 (Evade = Yes rows)
- k = 2 (Refund = Yes / No)

❖ With Laplace smoothing:

- $0 + 1/3 + 2 = 1/5 = 0.2$

Naïve Bayes Classifier: Comments

❖ Advantages

- Easy to implement
- Good results obtained in most of the cases

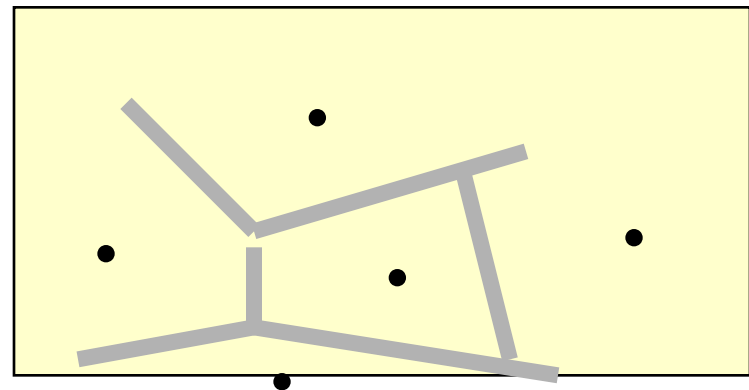
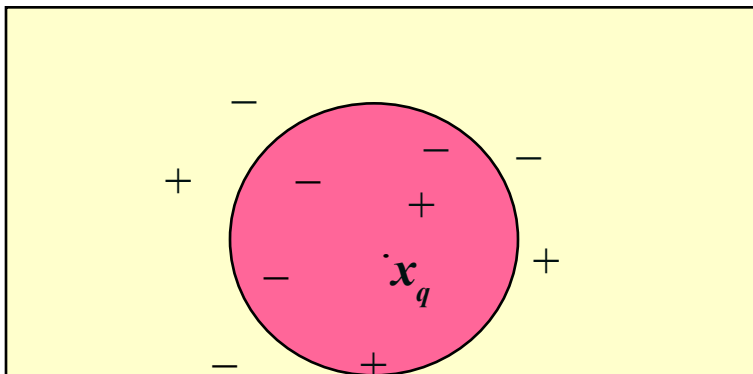
❖ Disadvantages

- Assumption: **class conditional independence**, therefore loss of accuracy
- Practically, **dependencies exist among variables**, e.g.:
 - Hospitals Patients Profile: age, family history, etc.
 - Symptoms: fever, cough etc.,
 - Disease: lung cancer, diabetes, etc.
- Dependencies among these **cannot be modeled by Naïve Bayes Classifier**

❖ How to deal with these dependencies? **Bayesian Belief Networks**

The k -Nearest Neighbor (k -NN) Algorithm

- ❖ All instances correspond to points in the n -D space
- ❖ The nearest neighbor are defined in terms of Euclidean distance, $\text{dist}(\mathbf{X}_1, \mathbf{X}_2)$
- ❖ Target function could be discrete- or real- valued
- ❖ For discrete-valued, k -NN returns the most common value (= majority vote) among the k training examples nearest to x_q
- ❖ Voronoi diagram: the decision surface induced by 1-NN for a typical set of training examples



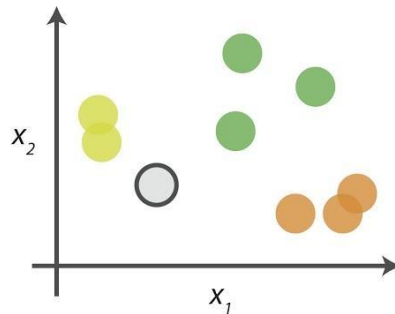
Discussion on the *k*-NN Algorithm

- ❖ *k*-NN for real-valued prediction for a given unknown tuple
 - Returns the mean values of the *k* nearest neighbors
- ❖ Distance-weighted nearest neighbor algorithm
 - Weight the contribution of each of the *k* neighbors according to their distance to the query x_q
 - Give greater weight to closer neighbors
- ❖ Robust to noisy data by averaging *k*-nearest neighbors
- ❖ Curse of dimensionality: distance between neighbors could be dominated by irrelevant attributes
 - To overcome it, axes stretch or elimination of the least relevant attributes
- ❖ Categorical (non-numerical) attributes, e.g., color, are difficult to handle

$$w \equiv \frac{1}{d(x_q, x_i)^2}$$

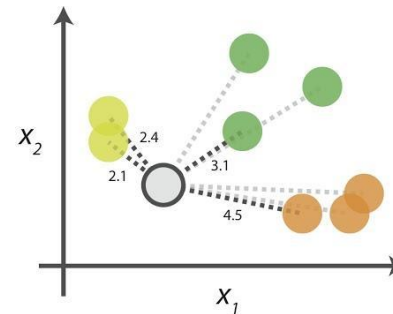
k-NN Algorithm

0. Look at the data



Say you want to classify the grey point into a class. Here, there are three potential classes - lime green, green and orange.

1. Calculate distances



Start by calculating the distances between the grey point and all other points.

2. Find neighbours

Point Distance			
		2.1	→ 1st NN
		2.4	→ 2nd NN
		3.1	→ 3rd NN
		4.5	→ 4th NN

Next, find the nearest neighbours by ranking points by increasing distance. The nearest neighbours (NNs) of the grey point are the ones closest in dataspace.

3. Vote on labels

Class	# of votes	
	2	➔ Class wins the vote! Point is therefore predicted to be of class .
	1	
	1	

Vote on the predicted class labels based on the classes of the k nearest neighbours. Here, the labels were predicted based on the k=3 nearest neighbours.

KNN Example

$X = (\text{Weight: 125 grams, Color Intensity: 4.5})$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Apply this for each row:

1. To Apple (180, 7):

$$\begin{aligned}\text{Distance} &= \sqrt{((180-125)^2 + (7-4.5)^2)} \\ &= \sqrt{(3025 + 6.25)} = \sqrt{3031.25} \approx 55.05\end{aligned}$$

2. To Apple (175, 6):

$$\text{Distance} \approx 50.31$$

3. To Banana (120, 4):

$$\text{Distance} \approx 5.10$$

4. To Banana (130, 5):

$$\text{Distance} \approx 5.10$$

5. To Grape (50, 9):

$$\text{Distance} \approx 84.55$$

Fruit	Weight (grams)	Color Intensity	Label
Apple	180	7	Apple
Apple	175	6	Apple
Banana	120	4	Banana
Banana	130	5	Banana
Grape	50	9	Grape

The top 3 neighbors (K=3) are: **Banana, Banana, Apple**
2 votes for Banana.

Label	Distance
Banana	5.10
Banana	5.10
Apple	50.31

Classifier Accuracy Measures

		Predicted	
		C_1	C_2
	Actual	C_1	C_2
	C_1	True positive	False negative
	C_2	False positive	True negative

classes	buy_computer = yes	buy_computer = no	total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
total	7366	2634	10000

	Predicted YES	Predicted NO
Actual YES	6954 (TP)	46 (FN)
Actual NO	412 (FP)	2588. (TN)

Total samples

$N. = TP + TN + FP + FN$

$N = 6954 + 2588 + 412 + 46 = 10000$

Evaluation Metrics

- *Accuracy measures the overall percentage of correct predictions made by the model.*
- Value significance (95.42%)
The model correctly predicts 95.42% of all cases.
- *Precision tells us how reliable the model's positive predictions are.*
- Value significance (94.41%)
When the model predicts buy_computer = yes, it is correct 94.41% of the time.

Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Calculation

$$\begin{aligned}\text{Accuracy} &= \frac{6954 + 2588}{10000} \\ &= \frac{9542}{10000} \\ &= 0.9542\end{aligned}$$

✓ Accuracy = 95.42%

Formula

$$\text{Precision} = \frac{TP}{TP + FP}$$

Calculation

$$\begin{aligned}\text{Precision} &= \frac{6954}{6954 + 412} \\ &= \frac{6954}{7366} \\ &= 0.9441\end{aligned}$$

✓ Precision ≈ 94.41%

Evaluation Metrics

- *Recall measures how many actual positive cases the model successfully detects.*
- Value significance (99.34%)
The model finds almost all real buyers, missing very few.
- *F1-score balances precision and recall into a single performance measure.*
- Value significance (96.81%)
The model achieves a strong balance between correctness and completeness of positive predictions.

Formula

$$\text{Recall} = \frac{TP}{TP + FN}$$

Calculation

$$\begin{aligned}\text{Recall} &= \frac{6954}{6954 + 46} \\ &= \frac{6954}{7000} \\ &= 0.9934\end{aligned}$$

✓ Recall $\approx$ 99.34%

Formula

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Substitute values

$$\begin{aligned}F1 &= \frac{2 \times (0.9441 \times 0.9934)}{0.9441 + 0.9934} \\ &= \frac{2 \times 0.9378}{1.9375} \\ &= 0.9681\end{aligned}$$

✓ F1-Score $\approx$ 96.81%

F1 Score / F-Score

- ❖ *How well the model makes correct positive predictions while also finding most of the real positives.*
- ❖ Range of F1-Score
 - $0 \leq F1 \leq 1$
 - **1** → Perfect model
 - **0** → Completely wrong model.

F1 Score Value	Interpretation
0.00 – 0.50	Poor performance
0.50 – 0.70	Fair / Needs improvement
0.70 – 0.85	Good model
0.85 – 0.95	Very good model
> 0.95	Excellent performance

Naïve Bayesian Classifier: Exercise

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

The given data belongs to which class (mammals/non-mammals)?

Use Naïve Bayesian classifier.

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

What Is Prediction?

- ❖ (Numerical) prediction is similar to classification
 - construct a model
 - use model to predict continuous or ordered value for a given input
- ❖ Prediction is different from classification
 - Classification refers to predict categorical class label
 - Prediction models continuous-valued functions
- ❖ Major method for prediction: regression
 - model the relationship between one or more *independent* or **predictor** variables and a *dependent* or **response** variable
- ❖ Regression analysis
 - Linear and multiple regression
 - Non-linear regression
 - Other regression methods: generalized linear model, Poisson regression, log-linear models, regression trees

Linear Regression

- ❖ Linear regression: involves a response variable y and a single predictor variable x

$$y = w_0 + w_1 x$$

where w_0 (y-intercept) and w_1 (slope) are regression coefficients

- ❖ Method of least squares: estimates the best-fitting straight line

$$w_1 = \frac{\sum_{i=1}^{|D|} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|} (x_i - \bar{x})^2} \quad w_0 = \bar{y} - w_1 \bar{x}$$

- ❖ Multiple linear regression: involves more than one predictor variable
 - Training data is of the form $(\mathbf{X}_1, y_1), (\mathbf{X}_2, y_2), \dots, (\mathbf{X}_{|D|}, y_{|D|})$
 - Ex. For 2-D data, we may have: $y = w_0 + w_1 x_1 + w_2 x_2$
 - Solvable by extension of least square method or using SAS, S-Plus
 - Many nonlinear functions can be transformed into the above

Example: Linear Regression

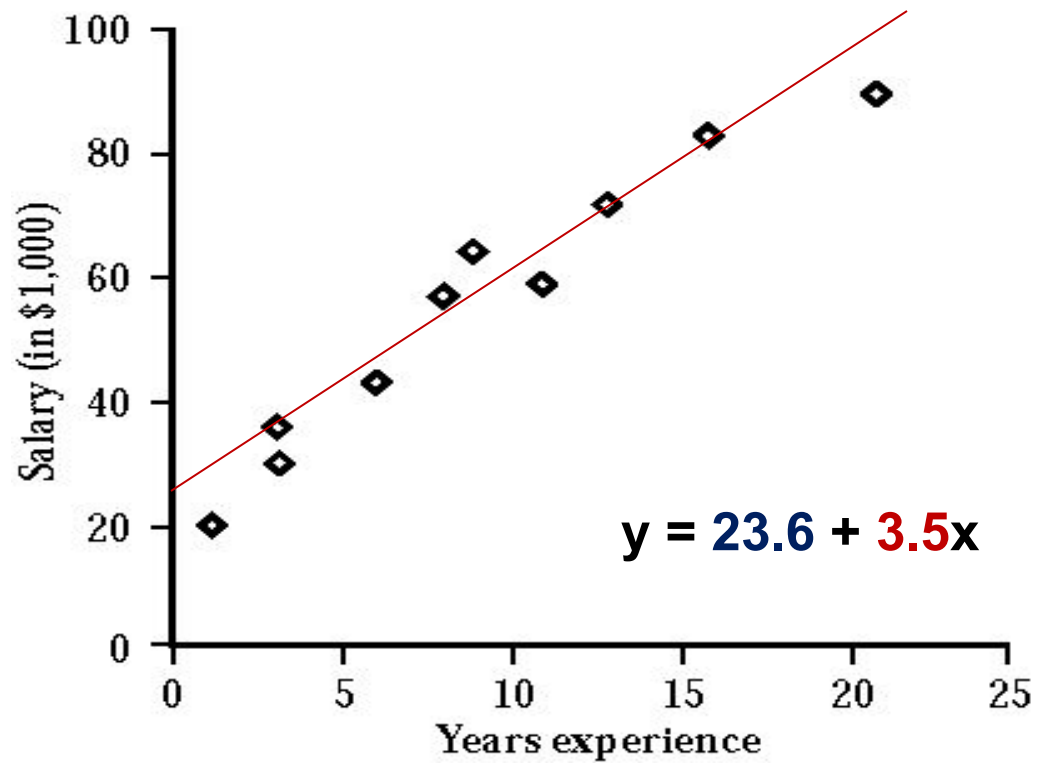
X (Years experience)	Y (Salary)
3	30K
8	57K
9	64K
13	75K
3	36K
6	43K
11	59K
21	90K
1	20K
16	83K

❖ Given data

$$\bar{x} = 9.1 \quad \bar{y} = 55.4$$

$$w_1 = \frac{(3-9.1)(30-55.4) + \dots + (16-9.1)(83-55.4)}{(3-9.1)^2 + \dots + (16-9.1)^2}$$

$$w_0 = 55.4 - (3.5)(9.1) = 23.6$$



Example: Linear Regression Analysis

	X (Years experience)	Y (Salary)	Y (Predicted) [$Y=23.6 + 3.5X$]
	3	30K	34.1K
	8	57K	51.6K
	9	64K	55.1K
	13	75K	69.1K
	3	36K	34.1K
	6	43K	44.6K
	11	59K	62.1K
	21	90K	97.1K
	1	20K	27.1K
	16	83K	79.6K
sum	91	557	
Avg	9.1	55.7	